

BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400058-India
Department of Computer Engineering

Group No.: 57

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Branch – CE

**TRANSITFLOW: A MULTI-CRITERIA, CROWD-AWARE ROUTE
COMPARISON SYSTEM FOR MUMBAI'S MULTI-MODAL PUBLIC
TRANSIT NETWORK**

Abstract

Mumbai commuters have no single platform to compare local train, metro, bus, and road options against personal priorities like time, cost, comfort, and predicted crowding. TransitFlow models the transit network – 112 stations spanning the Western and Central suburban rail lines and the Yellow, Red, and Aqua metro corridors – as a weighted, layered multigraph, with each station–mode pair represented as a distinct node and every service between two stations carrying independent time, cost, and comfort weights, joined by real walking interchanges.

Route discovery uses Yen's k-shortest-paths algorithm – Dijkstra's algorithm applied repeatedly with each found path's edges excluded – to surface genuinely distinct alternatives, such as different interchange choices, rather than a single best path per objective. Candidates are ranked by a normalized, priority-weighted combination of time, cost, and comfort, with comfort further adjusted by a rule-based crowd-prediction model driven by hour-of-day, weekday/weekend patterns, and per-line baselines, alongside line-aware rain penalties. A normalized relational database backs station, edge, interchange, and crowding data, with adjacency-list graph structures and priority queues enabling efficient computation, presented through an interactive, map-based frontend for transparent route comparison – demonstrating applied integration of graph theory, algorithm design, data structures, and database management in a real-world problem.

Components / Software List

Software / Technologies:

- Programming Languages: Python 3 (backend), JavaScript / JSX (frontend)
- Backend Framework: FastAPI (REST API), Uvicorn ASGI server
- Graph & Algorithms: NetworkX – weighted multigraph, Dijkstra's algorithm, Yen's k-shortest-paths
- Database: SQLite via SQLAlchemy ORM (station, edge, interchange, and crowding-rule tables)
- Frontend: React 19, Vite, Tailwind CSS, Framer Motion, Lenis (smooth scrolling)
- Mapping: MapLibre GL JS with react-map-gl (CARTO vector basemaps)
- Data Validation: Pydantic
- IDE / Tools: VS Code, Git & GitHub
- Hardware: Standard laptop/PC (min 8GB RAM) – no specialized hardware required

Core Subjects Mapped

- Discrete Structures and Graph Theory
- Design and Analysis of Algorithms
- Data Structures
- Database Management Systems

Faculty Guide: Prof. Pramod Bide

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